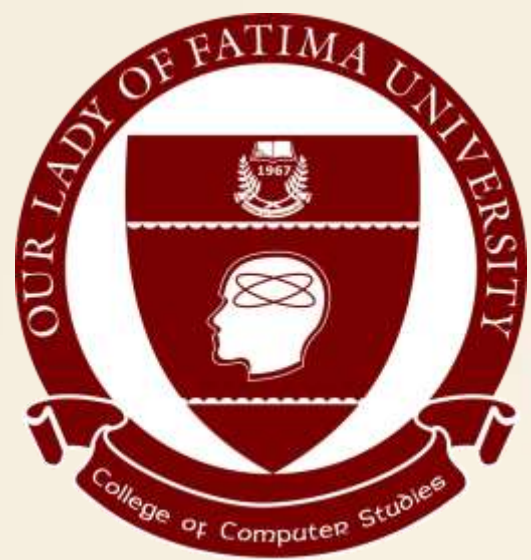




Module 12

Artificial Intelligence

Concepts

ITCL112 - INTRODUCTION TO COMPUTING W/ LAB

INTRODUCTION



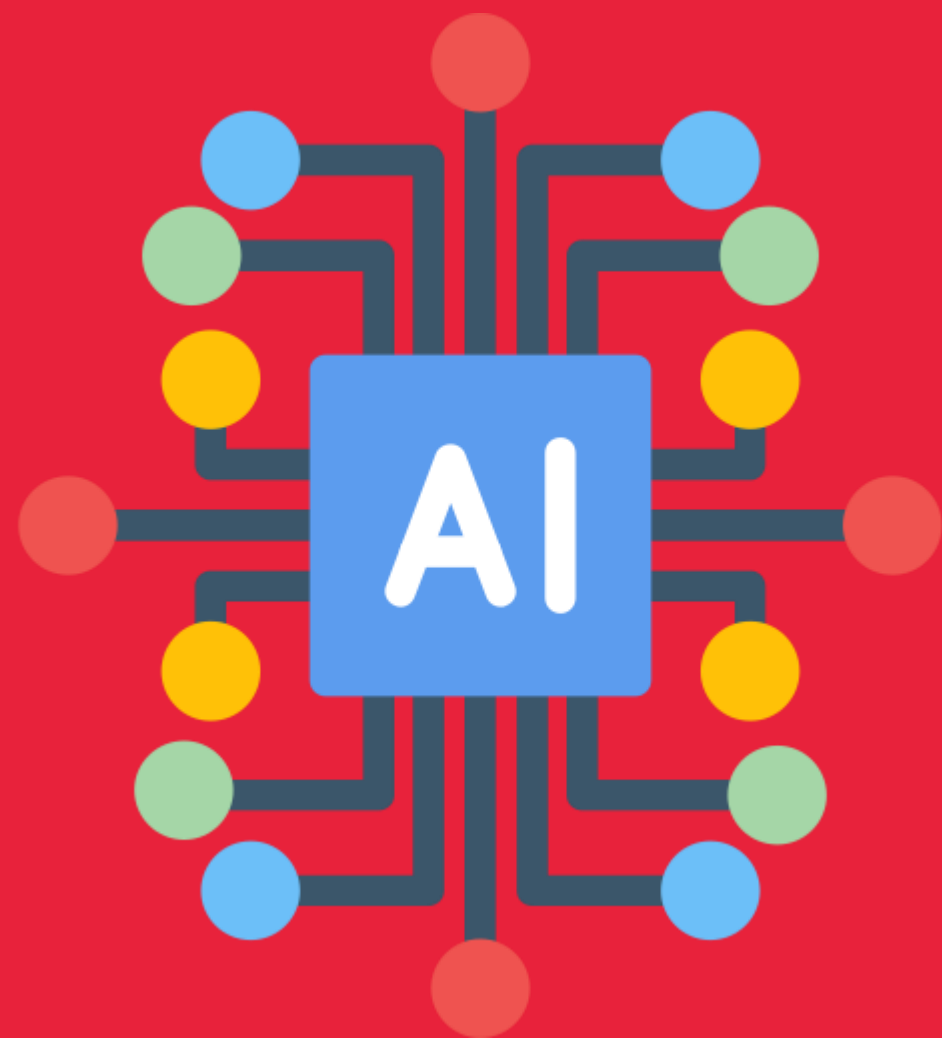
Artificial Intelligence (AI) is a branch of computer science dedicated to creating systems capable of simulating human intelligence to perform tasks such as visual perception, decision-making, and natural language processing. Unlike traditional software where programmers explicitly hard-code every logical rule, modern AI—specifically Machine Learning—uses statistical algorithms to process massive datasets, allowing the system to "learn" patterns and optimize its performance over time without human intervention. While the field aims toward the theoretical "General Intelligence" (AGI) seen in science fiction, today's technology is classified as "Narrow Intelligence" (ANI), meaning it is highly specialized to solve specific problems—like recognizing a face or recommending a movie—often more efficiently than a human, but lacking genuine consciousness or versatility.

MODULE 10

After this lesson, students should be able to:

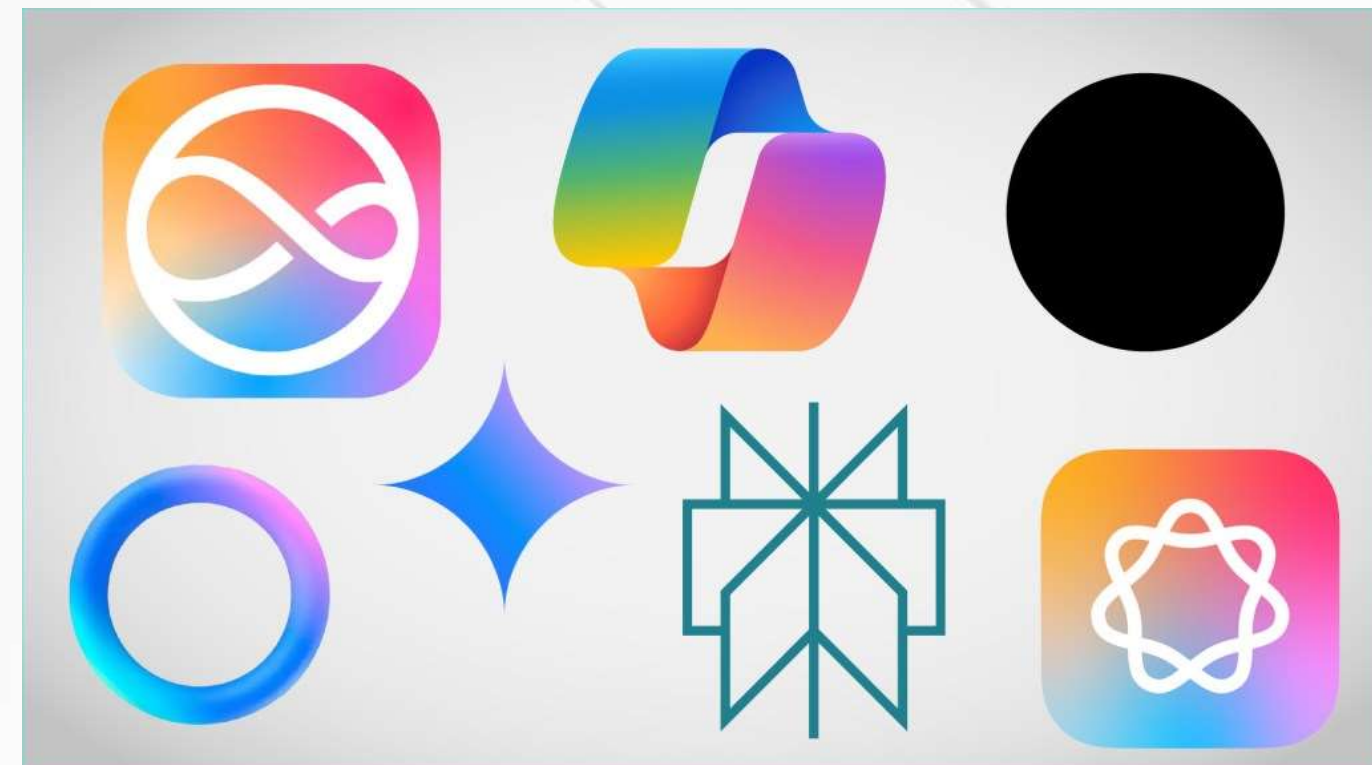
- 1. Understand how we moved from "programming rules" to "programming systems that learn."**
- 2. Technically classify AI based on how it processes data.**
- 3. Analyze AI as a deployed system within an IT infrastructure.**

What is Artificial Intelligence?



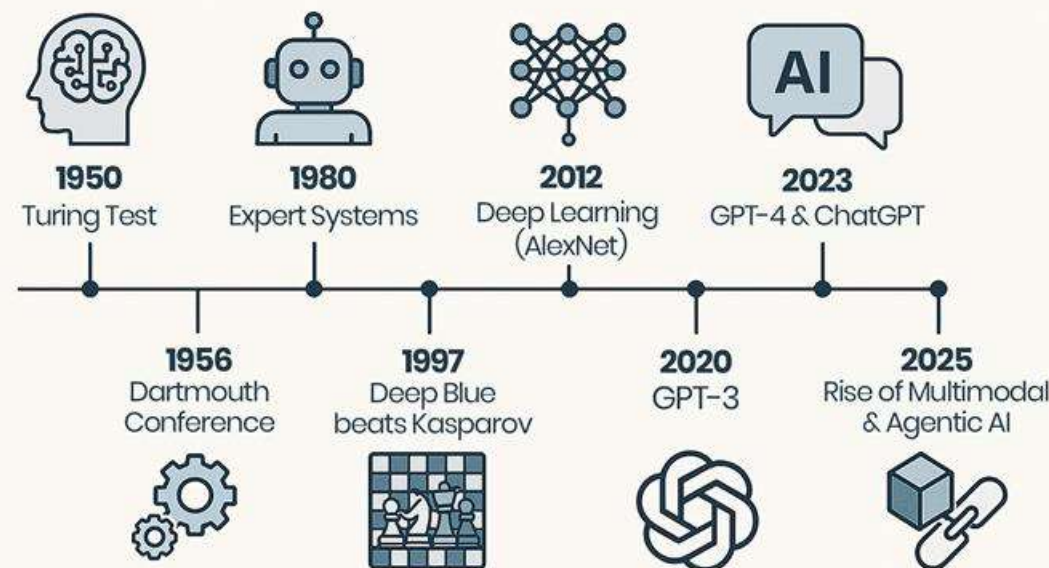
Artificial Intelligence (AI) is a branch of computer science dedicated to creating systems capable of performing tasks that typically require human intelligence. These tasks include recognizing speech, making decisions, translating languages, and identifying patterns.

In simple terms: Traditional programming is telling a computer what to do rule-by-rule. AI is creating a computer that can figure out what to do by learning from data.



History & The Shift in Computing Paradigms

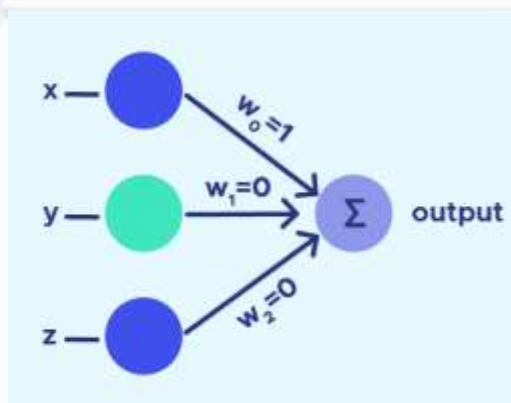
HISTORY OF ARTIFICIAL INTELLIGENCE



The Pre-History: The Concept of "Computability" (1930s-50s)

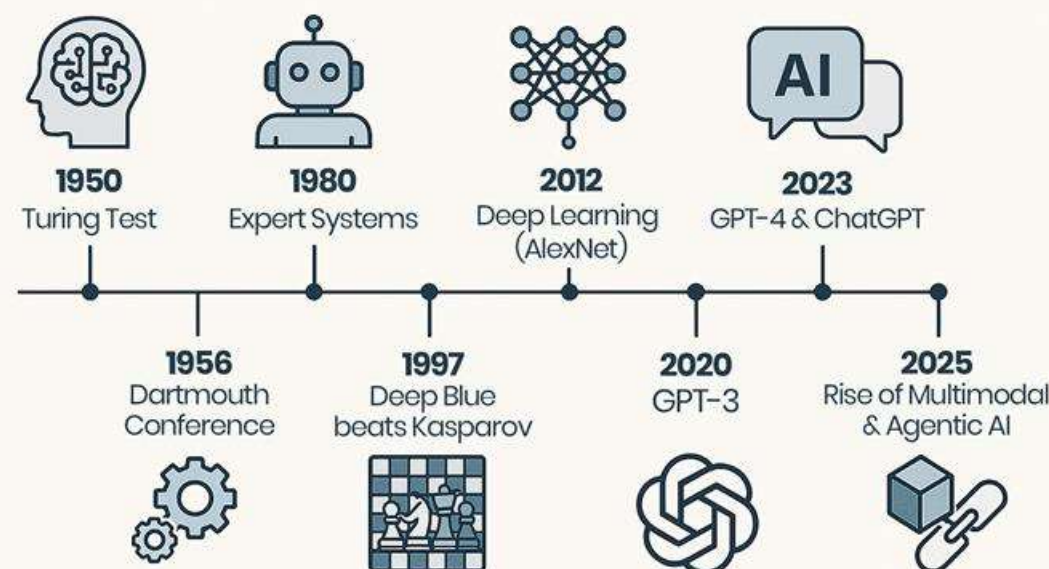
Before AI could exist, we needed to define what a computer could theoretically do.

- Alan Turing & The Turing Machine: Turing didn't just propose the "Turing Test"; he defined the mathematical model of a general-purpose computer. He proved that simple binary operations (0s and 1s) could represent any logical process.
- The Perceptron (1958): Frank Rosenblatt created the Perceptron, the first algorithmic model of a biological neuron.
- IT Concept: It took binary inputs, applied "weights" (importance), and produced an output. This is the great-grandfather of modern Deep Learning.



History & The Shift in Computing Paradigms

HISTORY OF ARTIFICIAL INTELLIGENCE



Symbolic AI / GOFAI (1960s-80s)

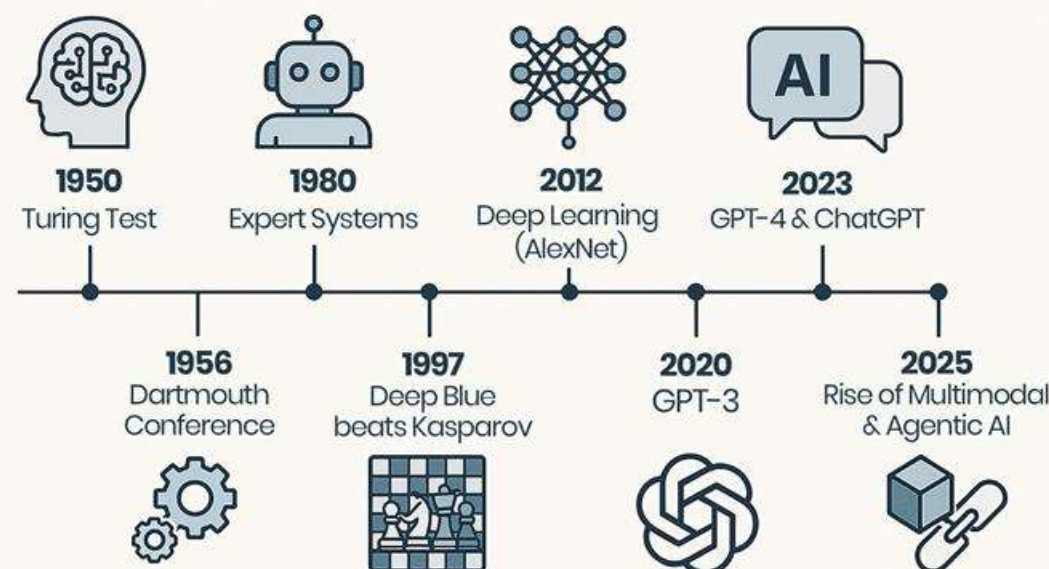
For decades, AI was Rule-Based. This is often called Good Old-Fashioned AI (GOFAI).

- How it worked: Programmers explicitly coded the logic using IF-THEN-ELSE statements.
- Example (Expert System): A medical diagnosis bot.
 - Code Logic: IF temp > 39 AND cough == True THEN diagnosis = 'Flu'
- The Failure (The Frame Problem): You cannot write enough rules to describe the real world. If you show a Rule-Based AI a picture of a cat upside down, it fails, because its rules say "ears must be on top." This rigidity led to the "AI Winters" (periods where funding stopped due to failure).



History & The Shift in Computing Paradigms

HISTORY OF ARTIFICIAL INTELLIGENCE



The Statistical Turn: Machine Learning (1990s-Present)

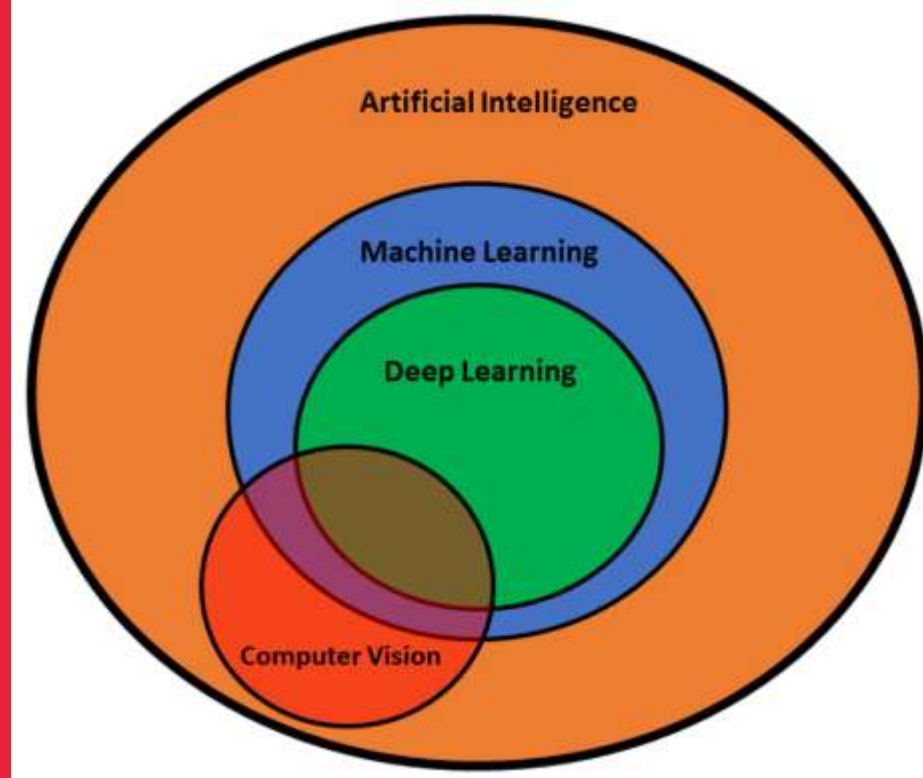
The paradigm shifted from Deductive Logic (top-down rules) to Inductive Logic (bottom-up data).

- Instead of telling the computer what a cat looks like, we feed it 10,000 images of cats and say, "Find the mathematical pattern."
- The mechanism: The computer adjusts its internal parameters to minimize the error rate between its guess and the correct answer.

In traditional programming, you input Rules and Data to get Answers. In Machine Learning, you input Data and Answers to get the Rules.



Architectures and Types of AI



Artificial Intelligence (AI):

The broad discipline of creating intelligent machines (includes the old rule-based systems).

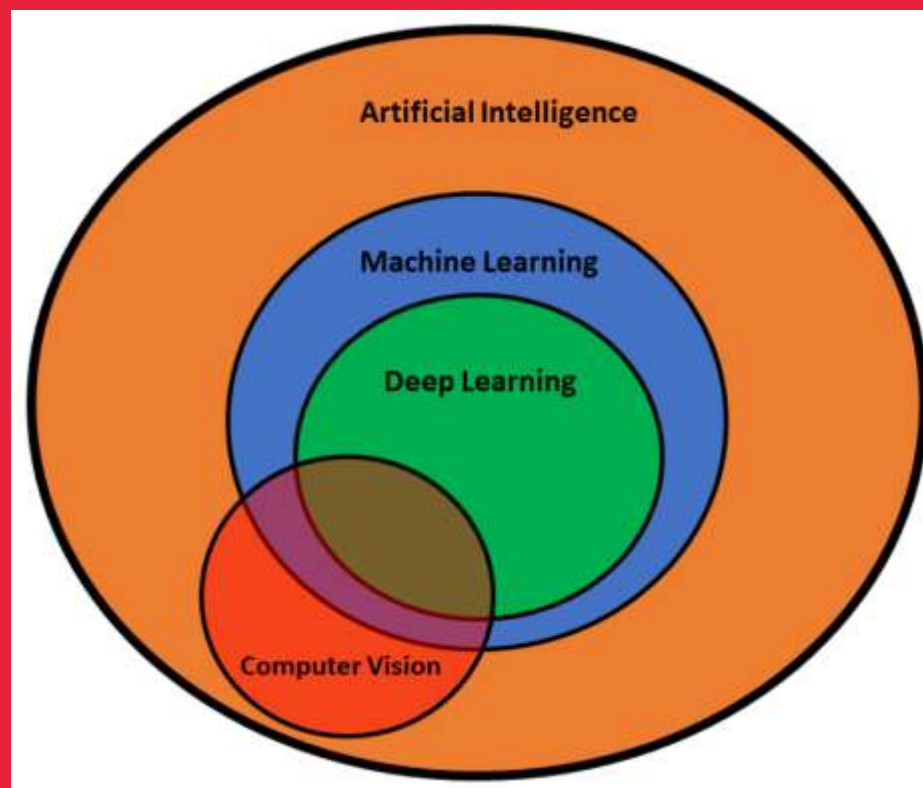
Machine Learning (ML):

A subset of AI using statistical methods. The algorithms parse data, learn from it, and make decisions.

Deep Learning (DL):

A specialized subset of ML inspired by the human brain structure.

Architectures and Types of AI



The "Neuron" in Deep Learning

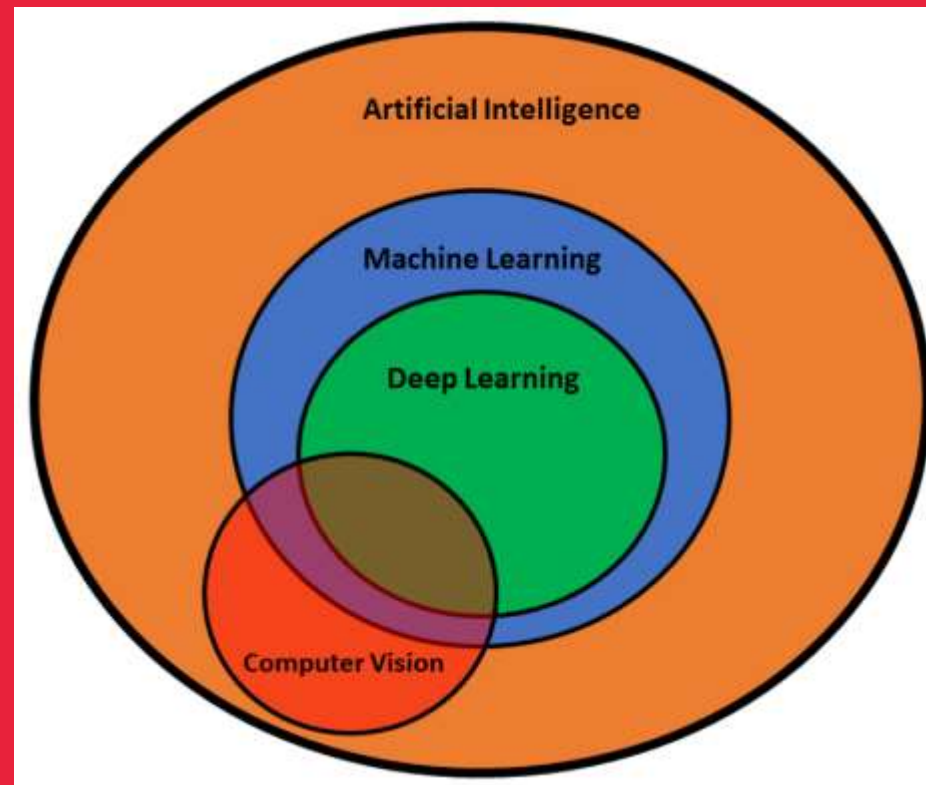
The Node (Neuron): A mathematical function.

The Formula: An output (y) is calculated by taking inputs (x), multiplying them by weights (w), adding a bias (b), and passing it through an activation function.

$$y = \text{Activation}(\sum (w \cdot x) + b)$$

Deep Neural Networks: "Deep" simply means there are many Hidden Layers of these neurons between the Input and Output. These layers extract features (edges -> shapes -> eyes -> faces).

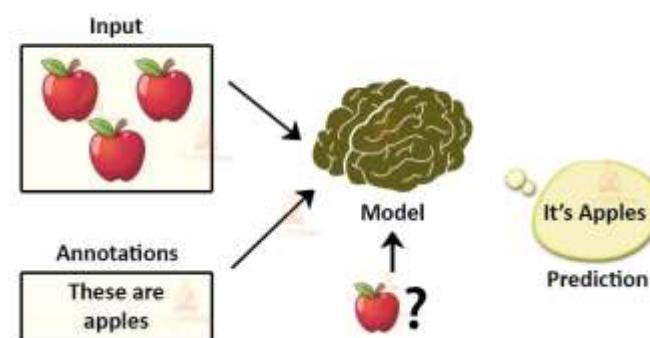
Architectures and Types of AI



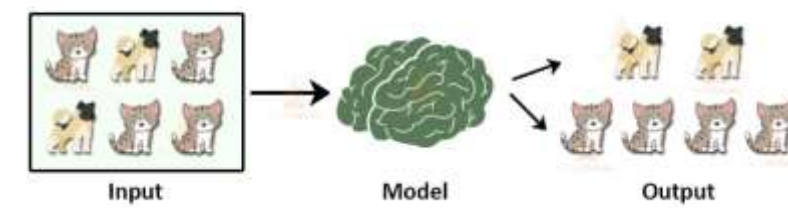
Types of Machine Learning (The "How")

Learning Type	Description	IT/Data Context
Supervised Learning	The model is trained on Labeled Data. We show it the image and the "tag."	Regression (Predicting a number, like house prices) or Classification (Spam vs. Not Spam).
Unsupervised Learning	The model is given Unlabeled Data and must find structure on its own.	Clustering (Grouping customers by purchasing behavior) or Anomaly Detection (Finding hackers in network logs).
Reinforcement Learning	An Agent learns by interacting with an Environment. It gets a "Reward" for good actions and "Penalty" for bad ones.	Used in Game AI (Dota 2 bots), Robotics walking, and optimized routing.

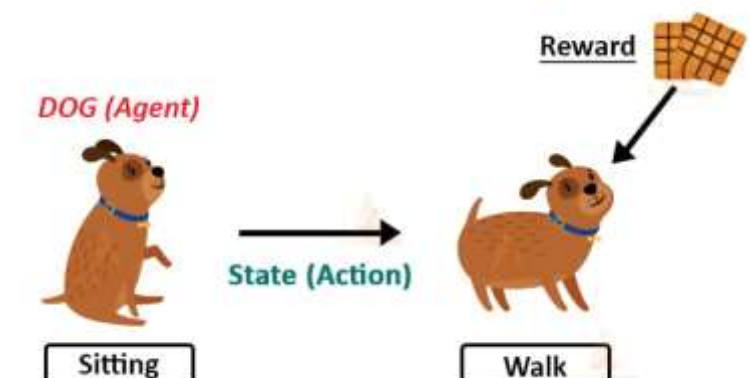
Supervised Learning in ML



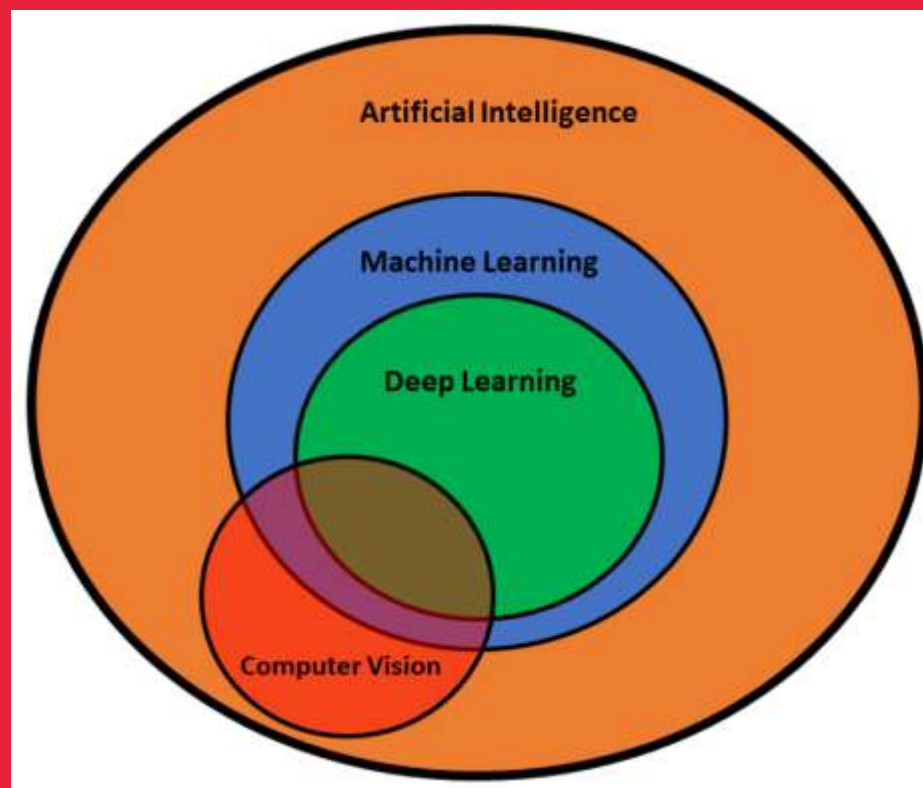
Unsupervised Learning in ML



Reinforcement Learning in ML



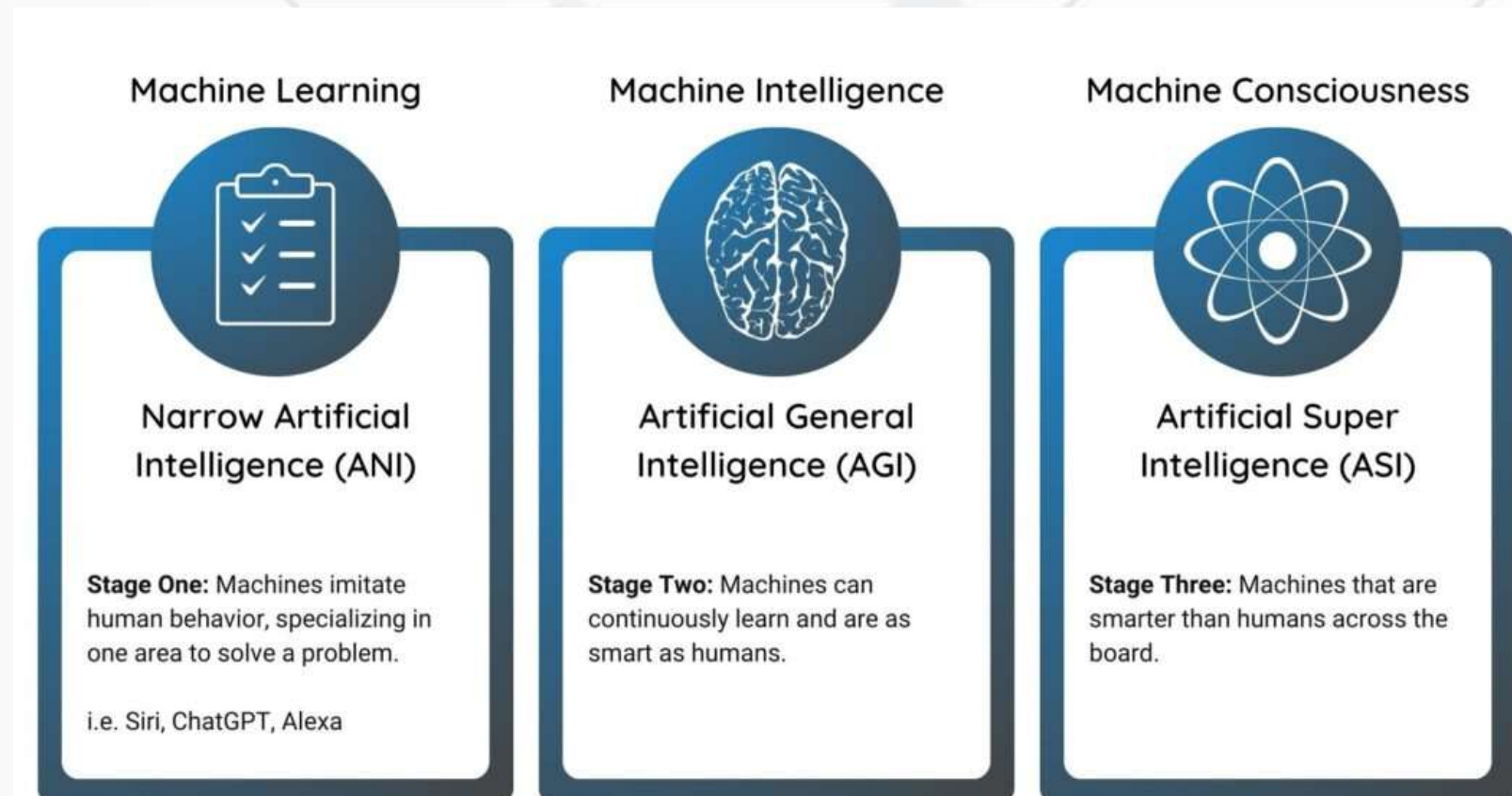
Architectures and Types of AI



Capabilities: ANI vs. AGI

ANI (Artificial Narrow Intelligence): Systems with Weak Generalization. They function efficiently only within the specific distribution of data they were trained on. (e.g., A poker bot cannot play Solitaire).

AGI (Artificial General Intelligence): Systems capable of Transfer Learning—taking logic learned in one domain and applying it to a completely new one without retraining.



Advantages and Disadvantages of AI



Advantages

Reduction in Human Error

"Human error" is born from fatigue or distraction. AI does not get tired. It is highly precise in calculation and data processing.

24/7 Availability

Humans need sleep and breaks; AI agents and chatbots can work endlessly without a drop in performance.

Handling Repetitive Tasks

AI can handle mundane tasks (like verifying documents or sorting emails), freeing humans to do creative work.

Risk Taking

AI robots can be used in hazardous environments, such as defusing bombs, exploring the ocean floor, or going to Mars.

Disadvantages

Job Displacement

As AI automates tasks, there is a significant risk of certain job roles becoming obsolete, particularly in manufacturing and data entry.

High Cost

Creating and maintaining complex AI systems requires massive hardware resources and energy consumption.

No Creativity or Emotion

While AI can mimic art, it lacks genuine emotional intelligence and the human capacity for original, lateral thinking.

Bias and Ethics

AI learns from human data. If the data is biased (racist, sexist, etc.), the AI will make biased decisions. This is known as "Algorithmic Bias."